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To: The Editor-in-Chief

Measurement and Control

Re: Submission of manuscript

“Commissioning-Calibrated GP-HOCBF Safety Filtering for Ultra-Supercritical Boiler-Turbine Control under Model Mismatch”

Dear Editor of *Measurement and Control*:

On behalf of the authors, I submit the manuscript entitled “*Commissioning-Calibrated GP-HOCBF Safety Filtering for Ultra-Supercritical Boiler-Turbine Control under Model Mismatch*” for consideration as an Original Research Article in *Measurement and Control*. We confirm that the manuscript is under consideration by *Measurement and Control* only. It has not been published or submitted elsewhere, in whole or in part, in any language. All authors have contributed to the work and approved its submission.

The manuscript presents RoCBF-SF, a commissioning-calibrated GP-HOCBF safety-filter overlay for a 660 MW ultra-supercritical boiler-turbine coordinated-control loop under model mismatch. The filter operates downstream of an existing controller, projects each proposed command through a real-time QP safety layer, and calibrates the robustness margin from commissioning residuals. The study combines a drift-only primary benchmark with field historian and controller-log evidence. The field evidence includes load-matched pre/post-retrofit comparisons. The controller logs report optimal QP status, no fallback activations, no manual-mode transitions, and no actuator-saturation events. They also show positive pressure, enthalpy, and power margins with millisecond-level solve times. This emphasis is consistent with recent *Measurement and Control* articles on uncertain industrial-process control, practical controller tuning, disturbance rejection, and response evidence from simulation, field testing, or experiments. The work is therefore aligned with the journal’s interest in industrial measurement, control implementation, and measured process evidence.

We believe that the manuscript will be of interest to readers concerned with deployable safety layers for industrial control loops, and we hope that you will consider it for publication in *Measurement and Control*.

Sincerely yours,

Dr. Zhuang Shao (on behalf of all authors)

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